Shivanshu Siyanwal

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RESEARCH INTERESTS

Quantum information and near-term quantum algorithms: variational quantum circuits and quantum machine learning for scientific data, quantum image representations, entanglement detection on NMR processors, linear-optical quantum networks and QKD, and hybrid quantum–classical implementations including FPGA inference.

EDUCATION

Indian Institute of Science Education and Research (IISER) Mohali **Punjab, India**
BS–MS (Dual Degree), Major: Physics; Minor: Data Science *Aug 2017 – May 2022*

- MS thesis: *Entanglement detection and gate optimization in the system of three and four NMR qubits using deep learning and machine learning.* Advisor: Prof. Kavita Dorai. NMR Quantum Computing Group.
- Thesis work became the co-first-author paper in *Quantum Information Processing* (2025) on ANN detection of multipartite entanglement.

RESEARCH EXPERIENCE

Centre for Development of Advanced Computing (C-DAC) **Noida, India**
Quantum Applications Engineer *Oct 2024 – Present*

- Co-developed quantum-assisted weather models (QGRU, QLSTM, QNN, VQC, QSVM) on ERA5 single-level data for Safdarjung, Delhi; equal-contribution preprint arXiv:2503.23408, and a related IEEE AIML-Systems 2025 paper.
- Compared quantum image encodings (NASS, QBIR, FTQR, FRQCI, and FRQI/QPIE/NEQR) inside circularly entangled QNNs for binary classification; equal-contribution paper in *The Journal of Supercomputing* (2026).
- Co-authored the FPGA deployment of a hybrid variational circuit for temperature forecasting on an AMD Alveo U55C (HLS, HBM; 291.8 µs inference, 6.7× versus a GPU baseline). IEEE TQCEBT 2026.

Institute for Advancing Intelligence, TCG CREST (PI: Dr. Arpita Maitra) **Kolkata, India**
DRDO Junior Research Fellow, Quantum Networks and Linear Optics *Mar 2024 – Oct 2024*

- Simulated DPS, COW, and twin-field QKD on linear-optical testbeds in OptiSystem and ANSYS, including loss, noise, and photonic channel impairments.
- Modelled intercept-resend and photon-number-splitting attacks and benchmarked resource-state and entanglement generation.

Institute for Advancing Intelligence, TCG CREST (PI: Dr. Arpita Maitra) **Kolkata, India**
Research Intern, Quantum Cryptography *Aug 2023 – Mar 2024*

- Built quantum circuits for an 8-bit AES (Mini-AES / S-AES) and Galois-field arithmetic; ran them on Qiskit and IBM Quantum hardware.
- Implemented a plaintext–ciphertext attack with Grover search and amplitude amplification.

Arqanum Technologies Pvt. Ltd. **Pune, India**
Quantum Computing Developer *Sep 2022 – Jul 2023*

- Implemented amplitude, phase, and angle encodings and Ising-Hamiltonian circuits; trained VQC, QSVM, and quantum Boltzmann machines for time series and financial models.
- Ran VQLS jobs on Amazon Braket and Rigetti hardware.
- Contributed code to MeitY–QCAL (Dicke-state simulation) and to an Arqanum–IIT Kharagpur project on a quantum cryptographic hash.

NMR Quantum Computing Group, IISER Mohali (PI: Prof. Kavita Dorai; co-supervisor: Prof. Arvind) **Punjab, India**
MS Research Student *May 2021 – Aug 2022*

- Designed an ANN that detects genuine multipartite entanglement and assigns one of six SLOCC classes from a reduced set of density-matrix elements (canonical form; 128 elements reduced to 18, then ANOVA-ranked).
- Validated on 30 experimental three-qubit states from an NMR processor; GME detection remained accurate with 4 features and SLOCC classification with 6. Equal contribution with V. Gulati.

PUBLICATIONS

Journal articles

- V. Gulati*, **S. Siyanwal***, Arvind, and K. Dorai. ANN-enhanced detection of multipartite entanglement in a three-qubit NMR quantum processor. *Quantum Information Processing* **24**(3), 74 (2025). Preprint: arXiv:2409.19739. *Equal contribution.
- A. Tiwari*, **S. Siyanwal***, S. Sakhuja, S. Kumar, and P. Tyagi. Quantum image representation-based quantum neural networks for binary classification. *The Journal of Supercomputing* **82** (2026). doi:10.1007/s11227-026-08621-3. *Equal contribution.

Conference papers

- N. Khan, **S. Siyanwal**, K. Islur, A. Tiwari, and H. Gupta. Hybrid variational quantum circuits based temperature forecasting with low-latency FPGA inference. *2026 International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT)*, pp. 1–6, IEEE (2026). IEEE Xplore:11681266.
- A. Tiwari, S. Kumar, P. Tyagi, **S. Siyanwal**, and S. Sakhuja. Quantum machine learning models for enhanced weather prediction. *2025 5th International Conference on AI-ML-Systems (AIMLSystems)*, pp. 446–451, IEEE (2026). IEEE Xplore:11330613.

Preprint

- S. Sakhuja*, **S. Siyanwal***, A. Tiwari, Britant, and S. Kashyap. Quantum-assisted machine learning models for enhanced weather prediction. arXiv:2503.23408 (2025). *Equal contribution.

Thesis

- **S. Siyanwal**. Entanglement detection and gate optimization in the system of three and four NMR qubits using deep learning and machine learning. BS–MS thesis, IISER Mohali (2022).

PRESENTATIONS

- Poster, Quantum Techniques in Machine Learning (QTML) 2024, University of Melbourne.
- Poster, Quantum Communication and Computation (QCC) 2025, IAI, TCG CREST.
- 12th International Conference on Mathematics and Computing (ICMC) 2026.

AWARDS AND FELLOWSHIPS

- INSPIRE-SHE Fellow, Department of Science and Technology, Government of India (2017–2022).
- DRDO Junior Research Fellowship, TCG CREST (2024).
- IBM Certified Associate Developer — Quantum Computation using Qiskit v0.2X (2026).
- Qiskit Global Summer School, Quantum Simulations, Advanced Level (2022).
- Qiskit Global Summer School, Quantum Machine Learning, Certificate of Excellence (2021).
- IBM Quantum Challenge Africa, Advanced Achievement (2021).
- TOEFL iBT 96 (2022).

TECHNICAL SKILLS

- **Quantum software:** Qiskit, PennyLane, Cirq, Perceval, Amazon Braket, CUDA-Q. Variational algorithms (VQE, VQC, VQLS, QAOA), QUBO formulations, quantum image encodings (FRQI, NEQR, QPIE, NASS, QBIR).

- **Methods:** NMR entanglement detection and SLOCC classification, QKD protocol simulation, quantum neural networks and quantum physics-informed models, tensor-network methods.
- **Scientific computing:** Python (NumPy, SciPy, QuTiP), PyTorch, TensorFlow; finite-difference discretisation (including Black–Scholes-type PDEs); Git and CI.
- **Platforms:** IBM Quantum, AWS Braket, Rigetti, AMD Alveo FPGA (HLS, HBM), OptiSystem, ANSYS.